SMARTBRIDGE
Let's Bridge the Gap

SKILL WALLET
A SmartBridge Product

Solution Requirements

Date	02 July 2026
Team ID	xxxxxx
Project Name	Smart Lender – Loan Eligibility Prediction Using Machine Learning
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Applicant Input	The system shall allow users to enter applicant details such as gender, marital status, education, self-employed status, applicant income, co-applicant income, loan amount, loan amount term, credit history, property area, and dependents for loan eligibility prediction.	High
2	Input Validation	The system shall validate all mandatory input fields, ensure valid data types, and prevent incomplete or invalid applicant information before performing loan eligibility prediction.	High

3	Machine Learning Prediction	The system shall preprocess applicant data and predict loan eligibility using the trained XGBoost machine learning model.	High
4	Prediction Result	The system shall display whether the loan application is **Approved** or **Rejected** along with the prediction result to the user.	High
5	Model Loading	The application shall load the trained machine learning model (**model.pkl**) and scaler (**scaler.pkl**) during application startup.	Medium
6	Data Preprocessing	The system shall preprocess user input by encoding categorical values and scaling numerical values before passing them to the machine learning model.	High
7	Error Handling	The system shall display appropriate validation and error messages for invalid or incomplete user inputs.	Medium
8	User Interface	The web application shall provide a responsive, simple, and easy-to-use interface that works across different devices and browsers.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Prediction should be generated quickly after user submits the application details.	Prediction generated within **2 seconds**.
2	Scalability	The application should support multiple users accessing the system simultaneously.	Supports at least **100 concurrent users** without significant performance degradation.
3	Security & Data Privacy	Applicant information should be processed securely and should not be permanently stored in the application.	Secure input handling and protection of user data.
4	Reliability & Availability	The application should consistently provide accurate loan eligibility predictions with minimal downtime.	**99% successful prediction rate** during testing and stable application availability.
5	Usability & Accessibility	The interface should be simple, responsive, and easy for users to understand and operate.	Users should complete a prediction in **three simple steps** with a clear interface.
6	Cloud Deployment	The application shall be deployed on the Render cloud platform for public accessibility.	Successfully deployed and accessible through the **Render deployment URL**.